

Renyu (Philip) Zhang

Updated in June 2026

CONTACT INFORMATION

Department of Decisions, Operations and Technology
CUHK Business School
The Chinese University of Hong Kong
Cheng Yu Tung Building
12 Chak Cheung Street
Sha Tin, N.T., Hong Kong, China

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Website: rphilipzhang.github.io/rphilipzhang

EDUCATION

Washington University in St. Louis, Olin Business School St. Louis, MO
Ph.D. Operations Management 05/2016

- Thesis: **Dynamic Pricing and Inventory Management: Theory and Applications**
- Advisors: Nan Yang, Fuqiang Zhang

Peking University, School of Mathematical Sciences Beijing, China
B.S. Mathematics 07/2011

ACADEMIC POSITION

The Chinese University of Hong Kong, CUHK Business School Hong Kong, China
Associate Professor (*with tenure*) Since 09/2022
Visiting Associate Professor 09/2021-08/2022

New York University Shanghai Shanghai, China
Visiting Scholar Since 07/2022
Assistant Professor of Operations Management 09/2016-06/2022
NYU Global Network Assistant Professor 07/2017-06/2022

New York University, Stern School of Business New York, NY
Visiting Assistant Professor of Operations Management 06/2017-06/2022

NON-ACADEMIC POSITION

Kuaishou Beijing, China
Economist & Tech Lead Since 12/2018

RESEARCH INTERESTS

Methodologies

Generative AI and large language models; machine learning and deep learning; causal inference and econometrics; reinforcement learning; data-driven optimization.

Applications

Platform and marketplace analytics; social network; AI agent; AI ethics; revenue management and pricing; inventory and supply chain management.

PEER-REFEREED PUBLICATIONS

1. **Dynamic Pricing and Inventory Management under Inventory-Dependent Demand**, *Operations Research*, **62**(5), 2014, 1077-1094 (with Nan Yang).

2. **Dynamic Pricing and Inventory Management under Fluctuating Procurement Costs**, *Manufacturing & Service Operations Management*, **17**(3), 2015, 321-334 (with Guang Xiao and Nan Yang).
3. **Dynamic Supply Risk Management with Signal-Based Forecast Multi-Sourcing, and Discretionary Selling**, *Production and Operations Management*, **26**(7), 2017, 1399-1415 (with Long Gao, Nan Yang, and Ting Luo).
4. **Trade-in Remanufacturing, Customer Purchasing Behavior, and Government Policy**, *Manufacturing & Service Operations Management*, **20**(4), 2018, 601-616 (with Fuqiang Zhang).
 - Finalist award in the *POMS College of Supply Chain Management 2015 Best Student Paper Competition*
5. **Comparative Statics Analysis of An Inventory Management Model with Dynamic Pricing, Market Environment Fluctuation, and Delayed Differentiation**, *Production and Operations Management*, **31**(1), 2022, 341-357 (with Nan Yang).
6. **Carpool Services for Ride-sharing Platforms: Price and Welfare Implications** *Naval Research Logistics*, **69**(4), 2022, 550-565 (with Xuan Wang).
7. **Competition and Coopetition for Two-sided Platforms**, *Production and Operations Management*, **31**(5), 2022, 1997-2014 (with Maxime Cohen).
 - Honorable Mention award in the *INFORMS Service Science Section 2017 Best Cluster Paper Award*
8. **Dynamic Pricing and Inventory Management in the Presence of Online Reviews**, *Production and Operations Management*, **31**(8), 2022, 3180-3097 (with Nan Yang).
9. **Cold Start to Improve Market Thickness on Online Advertising Platforms: Data-Driven Algorithms and Field Experiments**, *Management Science*, **69**(7), 2023, 3838-3860 (with Zikun Ye, Dennis Zhang, Heng Zhang, Xin Chen, and Zhiwei Xu).
 - Accepted at the 2022 *MSOM Service SIG Conference*
 - Finalist award in the *INFORMS Revenue Management and Pricing 2020 Student Paper Competition* (Primary Awardee: Zikun Ye)
10. **Data Aggregation and Demand Prediction**, *Operations Research*, **70**(5), 2022, 2597-2618 (with Maxime Cohen and Kevin Jiao).
 - Finalist award in the *INFORMS Data Mining Section 2019 Best Paper Award*
11. **The Impact of Social Nudges on User-Generated Content for Social Network Platforms**, *Management Science*, **69**(9), 2023, 5189-5208 (with Zhiyu Zeng, Hengchen Dai, Dennis Zhang, Heng Zhang, Max Shen, and Zhiwei Xu).
 - Finalist of the *MSOM Best Student Paper Competition 2023* (Primary Awardee: Zhiyu Zeng)
12. **Inventory Commitment and Monetary Compensation under Competition**, *Manufacturing & Service Operations Management*, **25**(6), 2023, 2142-2159 (with Junfei Lei, Fuqiang Zhang, and Yugang Yu).
13. **Content Promotion for Online Content Platforms with the Diffusion Effect**, *Manufacturing & Service Operations Management*, **26**(3), 2024, 1062-1081 (with Yunduan Lin, Mengxin Wang, Max Shen, and Heng Zhang).
 - Winner of the *INFORMS Social Media Analytics 2022 Best Student Paper Award* (Primary Awardee: Yunduan Lin)
 - Finalist of *INFORMS Minority Issues Forum Paper Competition 2024*
14. **Online Advertisement Allocation under Customer Choices and Algorithmic Fairness**, *Management Science*, **71**(1), 2025, 825-843 (with Xiaolong Li, Ying Rong, and Huan Zheng).
 - Accepted at the 2022 *MSOM Supply Chain Management SIG Conference*
15. **Deep Learning Based Causal Inference for Large-Scale Combinatorial Experiments: Theory and Empirical Evidence**, *Management Science*, forthcoming, 2025 (with Zikun Ye, Zhiqi Zhang, Dennis J. Zhang, and Heng Zhang).

- Accepted at the *ACM Conference on Economics and Computation (EC'23)*
- Second Prize of the *CSAMSE 2023 Best Paper Award*
- Winner of the *INFORMS Social Media Analytics 2023 Best Student Paper Award* (Primary Awardee: Zikun Ye)

16. **Nonprogressive Diffusion on Social Networks: Approximation and Applications**, *Management Science*, forthcoming, 2025 (with Yunduan Lin, Heng Zhang and Max Shen).
 - Accepted at the *ACM Conference on Economics and Computation (EC'24)*
 - Winner of the *POMS-HK 2023 Best Student Paper Award Competition* (Primary Awardee: Yunduan Lin)
 - Finalist of the *INFORMS Data Mining and Decision Analytics Workshop 2023 Best Paper Competition*
17. **Optimal Growth of a Two-Sided Platform with Heterogeneous Agents**, *Production and Operations Management*, forthcoming, 2025 (with Yixin Zhu, Hongfan Chen, and Sean X. Zhou).
 - Accepted at the 2024 *MSOM Technology, Innovation, and Entrepreneurship SIG Conference*

SELECTED WORKING PAPERS

18. **Dynamic Competition in Online Retailing: Implications of Network Effects** (with Xin Geng, Zheyu Jiang and Nan Yang).
19. **Attention as Investment: Algorithmic Management of Long Tail Creators** (with Qinlu Hu, Ao Huang and Ni (Nina) Huang).
 - Accepted at the 2024 *International Conference on Information Systems*
20. **The Value of AI-Generated Metadata for UGC Platforms: Evidence from a Large-scale Field Experiment** (with Xinyi Zhang, Chenshuo Sun and Khim-Yong Goh).
 - Accepted at the 2024 *Conference on Information Systems and Technology*
 - Accepted at the 2025 *MSOM Technology, Innovation, and Entrepreneurship SIG Conference*
 - Accepted at the *ACM Conference on Economics and Computation (EC'26)*
21. **Losing Face but Winning Fame: Social Comparison and Streamer Performance in Livestreaming Platforms** (with Qinlu Hu, Ao Huang and Ni (Nina) Huang).
22. **Synthesizing Evidence: Data-Pooling as a Tool for Treatment Selection in Online Experiments** (with Zhenkang Peng, Chengzhang Li and Ying Rong).
23. **The Effect of Gender-Aware Curation Algorithms on User Engagement in User-Generated Content Platforms** (with Xinyi Zhang, Khim-Yong Goh, and Chenshuo Sun).
 - Accepted at the 2025 *Conference on Information Systems and Technology*
24. **Improving Behavioral Alignment in LLM Social Simulations via Context Formation and Navigation** (with Letian Kong, Qianran (Jenny) Jin).
 - Accepted at the 2025 *Conference on Information Systems and Technology*

BOOKS

25. *Demand Prediction in Retail - A Practical Guide to Leverage Data and Predictive Analytics*, Springer Series in Supply Chain Management, 2022 (with Maxime C. Cohen, Paul-Emile Gras, and Arthur Pentecoste).

SEMINARS AND CONFERENCE TALKS

- 2026 FDU School of Management, SUFE Business School, CAS AMSS, CKGSB, Miami Herbert, NYU Stern, Deming Workshop on Experimentation for Decision-Making, University of Macau Faculty of Business Administration
- 2025 INFORMS, INFORMS WDS, ORSC Online Forum, PKU Guanghua, CSAMSE, CEIBS, MSOM SIG, RUC Renmin Business School, Beijing Institute of Technology School of Economics and Management, XJTU School of Management, ZJU School of Management
- 2024 ICIS, NYU Alumni Association, SJTU Antai, All Saints Music, HKUST-USC Marshall Confer-

ence on Digital Transformation

- 2023 ASU Carey, McGill, DataFun Causal Inference Online Summit, INFORMS, INFORMS WDS, ASU Carey, Fudan University School of Management, HKU Business School, Southeast University School of Economics and Management, CSAMSE, EC, RMP, Kuaishou, Miami Herbert, Tencent, NUS Business School, CUHK-SZ SME, Young Statistician Forum [[Video link](#)]
- 2022 CityU HK College of Business, Club Bear Online Seminar, NYU Shanghai, Academia Café Online Workshop, Tsinghua University SEM, PKU Guanghua, Analytics for X, CUHK Business School GRF/ECS Info-Sharing Session, NYU Stern, CUHK PhD Recruitment Workshop, RMP, West China Medical Center, Sichuan University School of Business, NUS Business School, NUS ISEM, U of Toronto Rotman
- 2021 NJU School of Management and Engineering, INFORMS, CSAMSE, Tsinghua University IE, MSOM, HKUST Business School
- 2020 CUHK Business School, DataFun Talk, Tsinghua University SEM, CUHK Shenzhen School of Data Science, HKU Business School, INFORMS, Sun Yat-sen University Lingnan College, Fudan University School of Management, NSFC PI Meeting, Sichuan University School of Business, NUS IORA [[Video link](#)], SJTU Antai, NYU Stern, Club Bear Online Seminar [[Video link](#)]
- 2019 INFORMS, UIC College of Business Administration, Data-Driven and Innovative Operations Workshop, POMS HK, Young Scholars' Forum on Management Science
- 2018 PKU Guanghua, Sun Yat-sen University Lingnan College, JHU Carey, NYU Stern, Operations Management under "Internet+" Environment Workshop, MSOM, RMP, Revenue Management Workshop, POMS, POMS HK, CUHK Business School
- 2017 Young Scholars' Forum on Management Science, Sichuan University School of Business, WUSTL Olin, INFORMS, HK PolyU Faculty of Business, CityU HK College of Business, HKUST Business School, NYU Stern, MSOM, SMU Lee Kong Chian School of Business, NUS Business School, USTC School of Management, POMS HK
- 2016 Southeast University School of Economics and Management, Tsinghua University SEM, SJTU Antai, U of Toronto Rotman, INFORMS, NJU School of Management and Engineering, Southwest University of Finance and Economics School of Business Administration, Southwest Jiaotong University School of Economics and Management, SUFE School of Business Administration, CSAMSE, RMP, POMS, NYU Shanghai, UTD Jindal, NYU Stern, CUHK SEEM, HKUST Business School, HKUST IELM, HK PolyU Faculty of Business, CityU HK College of Business
- 2015 DSI, INFORMS, MSOM, POMS
- 2014 INFORMS, MSOM
- 2013 INFORMS
- 2012 INFORMS

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| HONORS, AWARDS,
AND GRANTS | 1. Alibaba Cloud, Joint Research Program, PI, HKD 435,405. | 03/2026-03/2028 |
| | 2. Hong Kong Research Grants Council, Research Fellow Scheme, [Link], PI, HKD 5,465,160. | 01/2026-12/2030 |
| | 3. CUHK Business School Research Excellence Award 2024-2025. | 06/2025 |
| | 4. Didi Marketplace Technology, Joint Research Program, PI, RMB 700,000. | 07/2025-06/2026 |
| | 5. Finalist, <i>INFORMS Minority Issues Forum Paper Competition</i> | 10/2024 |
| | 6. National Natural Science Foundation of China, Excellent Young Scientist Program 72422004, PI, RMB 2,000,000. | 01/2025-12/2027 |
| | 7. Hong Kong Research Grants Council, General Research Fund 14503224, PI, HKD 886,865. | 01/2025-12/2027 |

8. Finalist, *INFORMS Data Mining and Decision Analytics Workshop Best Paper Competition* 10/2023
9. Finalist, *MSOM Best Student Paper Competition* (Primary Awardee: Zhiyu Zeng) 10/2023
10. Winner, *INFORMS Social Media Analytics Best Student Paper Award* (Primary Awardee: Zikun Ye). 10/2023
11. Second Prize, *CSAMSE Best Paper Award*. 07/2023
12. Hong Kong Research Grants Council, General Research Fund 14504123, PI, HKD 444,066. 01/2024-12/2025
13. Tencent Collaboration with University Fund, PI, RMB 332,310. 12/2022
14. Winner, *POMS-HK Best Student Paper Award* (Primary Awardee: Yunduan Lin). 01/2023
15. Winner, *INFORMS Social Media Analytics Best Student Paper Award* (Primary Awardee: Yunduan Lin). 10/2022
16. CUHK 'Improvement on Competitiveness in Hiring New Faculties' Funding Scheme, PI, HKD 1,500,000. 07/2022
17. Hong Kong Research Grants Council, General Research Fund 14502722, PI, HKD 629,325. 09/2022-08/2025
18. Finalist, *INFORMS Revenue Management and Pricing Student Paper Competition* (Primary Awardee: Zikun Ye). 10/2020
19. Finalist, *INFORMS Data Mining Section Best Paper Award*. 10/2019
20. Shanghai Eastern Scholar, Young Scientist Program QD2018053, PI, RMB 200,000/Year.01/2019-12/2021
21. National Natural Science Foundation of China, Young Scientist Program 71802133, PI, RMB 180,000. 01/2019-12/2021
22. Honorable Mention, *INFORMS Service Science Best Cluster Paper Award*. 10/2017
23. Shanghai Pujiang Talent Program 17PJC074, PI, RMB 100,000. 09/2017-08/2019
24. Finalist Award, *POMS College of Supply Chain Management Best Student Paper Competition*. 05/2015
25. First Prize Award, Chinese Mathematics Olympiad 10/2006

POST-DOCTORAL FELLOW ADVISING

1. Zhenkang Peng (CUHK 2024-2026, SJTU Antai PhD 2024).
2. Yushan Zhou (CUHK 2025-2027, DUT School of Management PhD 2025).

PHD STUDENT ADVISING

1. Kevin Jiao (Dissertation Committee, NYU Stern 2019, Initial Placement: FINRA).
2. Xiaolong Li (SJTU Antai 2019, Initial Placement: Post-Doc at NUS IORA, then Assistant Professor at Durham U Business School).
3. Zikun Ye (UIUC IESE 2023, Initial Placement: Assistant Professor of Marketing, UW Foster).
4. Zhiyu Zeng (Dissertation Committee, Tsinghua IE 2023, Initial Placement: Post-Doc at WUSTL Olin, then Assistant Professor at SJTU Antai).
5. Junfei Lei (UW Seattle Foster 2023, Initial Placement: Post-Doc at INSEAD).
6. Yunduan Lin (Berkeley Civil Engineering 2024, Initial Placement: Assistant Professor, CUHK Business School).
7. Zhiqi Zhang (WUSTL Olin 2026, Initial Placement: Assistant Professor, Notre Dame Mendoza).
8. Zheyu Jiang (Dissertation Committee, Miami Herbert 2023, Initial Placement: Assistant Profes-

- sor, NJU School of Management and Engineering).
9. Zihan Zhao (Dissertation Committee, WUSTL Olin Current Student).
 10. Yixin Zhu (Dissertation Committee, CUHK Business School 2025, Initial Placement: SUFE Information Management and Engineering).
 11. Qinlu Hu (Advisor, CUHK Business School 2026, Initial Placement: Nova SBE).
 12. Xinyu Xu (Advisor, CUHK Business School Current Student).
 13. Shu Zhang (Advisor, CUHK Business School Current Student).
 14. Xinyi Zhang (NUS School of Computing 2026, Initial Placement: Post-Doc at NUS).
 15. Yifan Ren (Advisor, CUHK Business School Current Student).
 16. Letian Kong (CUHK Business School Current Student).
 17. Zhengyun Yu (Advisor, CUHK Business School Current Student).
 18. Keyu Chen (Advisor, CUHK Business School Current Student).
 19. Ao Huang (University of Miami Herbert Current Student).
 20. Ziyu Xiong (PKU Guanghua Current Student).

TEACHING EXPERIENCES

The Chinese University of Hong Kong, CUHK Business School

Coordinator

- Artificial Intelligence Empowered Business (Undergraduate)
 - Spring 2026

Instructor

- Artificial Intelligence for Business Research (PhD)
 - Spring 2026 (**GitHub Link**), Spring 2025 (5.91/6, **GitHub Link**), Spring 2024 (5.91/6, **GitHub Link**)
- Artificial Intelligence and Business Analytics (EMBA - Chinese)
 - Summer 2026
- Artificial Intelligence and Big Data Strategies (EMBA - Chinese)
 - Summer 2025 (5.71/6), Summer 2024 (5.69/6)
- Artificial Intelligence for Business (EMBA)
 - Winter 2025, Fall 2024 (5.07/6), Winter 2023 (5.73/6)
- Intelligent Business with Data Science (EMBA)
 - Winter 2022 (5.18/6)
- Advanced Business Analytics Practicum (MS in Business Analytics)
 - Spring 2026, Winter 2024 (5.53/6)
- Machine Learning for Business (I) (MS in Business Analytics, MS in Information Technology Management)
 - Winter 2025
- Business Intelligence Techniques and Applications (MS in Business Analytics)
 - Winter 2024 (5.88/6, 5.89/6), Winter 2023 (6/6, 5.95/6), Winter 2022 (5.94/6, 5.89/6), Winter 2021 (5.95/6, 5.94/6)
- Web Analytics and Intelligence (MS in Business Analytics)

- Winter 2023 (5.71/6)
- Business Analytics (Undergraduate)
 - Spring 2022 (5.33/6)

Teaching Grants and Awards

- CUHK TDLEG, “Integrating AI Competency, Ethics, and Innovation into Business Education,” PI, HKD 425,000
 - 01/2025-12/2025
- CUHK TDLEG, “AI-Driven Transformation of Business Education: Bridging Market Demands and University Education,” PI, HKD 420,440
 - 01/2026-12/2027
- CUHK Vice-Chancellor’s Exemplary Teaching Award
 - 2025
- Faculty Teaching Excellence Award
 - 2024-25
- Faculty Teaching Merit Award
 - 2023-2024, 2022-2023

New York University Shanghai

Instructor

- Machine Learning for Business (MS in Data Analytics and Business Computing)
 - Summer 2021 (4.3/5), Summer 2020 (4.8/5)
- Business Analytics (Undergraduate)
 - Spring 2021 (4.9/5, 4.9/5), Spring 2019 (4.2/5, 4.8/5), Spring 2018 (4.4/5, 4.7/5)
- Operations Management (Undergraduate)
 - Spring 2019 (4.6/5), Fall 2017 (3.8/5), Spring 2017 (4.0/5), Fall 2016 (4.6/5)

Business Honors Thesis Advisor

- Ziheng Wang (2021), Yiyi Zhang (2018), Yujia Ni (2018).

Shanghai Jiao Tong University

Instructor

- Artificial Intelligence for Business Research (PhD)
 - Summer 2024

Kuaishou

Internal Instructor

- Introduction to Economics 08/2022-12/2022
- Data Driven Optimization 05/2020
- Operations Analytics 12/2018-05/2019

PROFESSIONAL
SERVICES

Journal Editor

<i>Manufacturing & Service Operations Management</i> , Associate Editor.	<i>Since 03/2026</i>
<i>Operations Research</i> , Associate Editor.	<i>Since 01/2025</i>
<i>Production and Operations Management</i> , Senior Editor.	<i>Since 09/2022</i>
<i>Naval Research Logistics</i> , Associate Editor.	<i>Since 08/2022</i>

Journal, Conference, and Grant Referee

Management Science, Operations Research, Manufacturing & Service Operations Management, MSOM SIG Conference, Marketing Science, Information Systems Research, NeurIPS, ICML, ICLR, AAAI, Mathematics of Operations Research, Production and Operations Management, Journal of Operations Management, Transportation Science, Transportation Research: Part B, ISTTT, Service Science, Naval Research Logistics, Decision Sciences, European Journal of Operations Research, Omega, Journal of Management Science and Engineering, Applied Mathematical Modeling, Operational Research: An International Journal, HK RGC GRF/ECS Scheme, Singapore SSHR-TG.

Service Awards

Manufacturing & Service Operations Management Meritorious Service Award 2018, 2019, 2022, 2023, 2024

Tenure/Promotion Case External Reviewer

SUFE College of Business

Professional Society Board

INFORMS Service Science Section Communication Editor.	<i>Since 01/2023</i>
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Competition Judge

POMS-HK Best Student Paper Competition Co-Chair (2025).
INFORMS Service Science Best Cluster Paper Competition Co-Chair (2024, 2025).
INFORMS Service Science Student Paper Competition Co-Chair (2022, 2023).
MSOM Student Paper Competition (2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025).
RMP Student Paper Competition (2025).
POMS College of Supply Chain Management Student Paper Competition (2020).
CSAMSE Best Paper Competition (2021, 2022, 2023, 2024, 2025).
POMS International Conference China Best Paper Award (2022).
ISCOM Best Paper Competition (2021)

Conference Panelist, Committee, and Session Chair

Organizer:

- INFORMS Service Science Online Forum Series.

Panelist:

- INFORMS Service Science Doctoral Consortium 2024, CSAMSE 2023 Young Scholar Colloquium.

Session Chair:

- POMS-HK 2024, CSAMSE 2023, POMS-HK 2023, INFORMS 2020, INFORMS 2018, POMS 2018, POMS-HK 2018, POMS-HK 2017, CSAMSE 2016.

Sponsorships and Exhibits Committee member:

- POMS 2016.

External Program Reviewer

CUHKSZ School of Management and Economics MSBA Program.

*Since 01/2023***Professional Organization Membership**

INFORMS, MSOM, ISS, RMP, Service Science, POMS.

Internal ServicesCUHK Business School Education and Student Experience Sub-group for Strategic Plan 2026–30.
Since 11/2025

CUHK Business School Teaching and Education AI Task Force Chair. 11/2025-04/2026

CUHK Business School AI Research Hub Associate Director and Convenor. *Since 11/2024*

CUHK Business School Operations Management Faculty Recruiting Committee. 2023-2025

CUHK-CUHKSZ Joint Operations Management Workshop Organizer. 11/2023

CUHK Business School DOT Department Graduate Panel. 09/2023-07/2025

CUHK Business School Faculty Impact Review Panel. *Since 01/2023*CUHK Business School Faculty RAE Panel. *Since 01/2023*CUHK Business School, DOT Department Computing Resources Coordinator. *Since 02/2024*CUHK Business School Operations Management PhD Reading Session Organizer. *Since 09/2022*

CUHK Business School DSE Department Name Change Task Force. 09/2022-06/2023

CUHK Business School IBBA Business Analytics Concentration Advisor. *Since 09/2022*

CUHK Business School Curriculum Committee. 09/2021-08/2022

NYU Shanghai Business Honors Program Advisor and Lecturer. 09/2017-05/2021

NYU Shanghai Dean's Undergraduate Research Fund Research Advisor. 03/2018-08/2022

NYU Shanghai Candidate Weekend Instructor and Interviewer. 03/2018

NYU Shanghai MSDABC Program Admission Event Guest Speaker. 08/2020

NYU Shanghai Operations Management Reading Group Organizer. 03/2017-06/2019

NYU Shanghai Housing Subcommittee Member. 11/2016-05/2019

NYU Shanghai Business Research Seminar Coordinator. 09/2016-08/2017

PERSONAL

Chinese citizen, Christian, married with one child.